

LYCANS AERODESIGN TEAM

Training Task 01

GitHub Setup & Fixed-Wing Competition Research

Deadline: Aug 18, 2026 - 11:59 PM
Submission: GitHub Repository

1 Objective

This task introduces the workflow that will be used for future Lycans training assignments.
You will practice:

- Creating and organizing your own GitHub repository
- Following a standardized repository naming convention
- Conducting technical research
- Writing a structured technical document using LaTeX
- Using Overleaf to compile your document
- Submitting your work through GitHub

2 Part 1 — Create Your GitHub Repository

Create your personal training repository using the official Lycans training template.
This repository will be your **main training repository** throughout the program.
You will use the **same repository for all future assignments**.

2.1 Template

<https://github.com/Lycans-Training-27/lycans-training-template>

Click:

Use this template → **Create a new repository**

2.2 Repository Owner

Your repository must be created under:

Lycans-Training-27

2.3 Repository Name

Use the following naming convention:

[ApplicantCode] – [Name]

Examples:

- E001–Ahmed
- M004–Mohamed
- E012–Omar

Use the applicant code assigned to you.

2.4 Repository Visibility

Your repository must be:

Private

Do not create your assignment repository under your personal GitHub account.

2.5 Repository Structure

Your repository will be used for **all training assignments**.

Each assignment must have its own folder.

The provided `assignment.tex` file is a **reference template**.

Do not edit, rename, or delete the template file.

For each assignment, create a new folder and create a new `.tex` file inside it.

Your repository should eventually look like:

```
repository/
|
+-- assignment.tex          <- Reference template (DO NOT EDIT)
|
+-- images/
|   +-- lycans-logo.png
|
+-- task-01/
|   +-- task-01.tex
|
+-- task-02/
|   +-- task-02.tex
|
+-- task-03/
|   +-- task-03.tex
|
+-- ...
```

The template is provided so that you can refer back to the formatting, commands, examples, and structure when writing future assignments.

Each assignment should be a separate document.

2.6 Assignment Folder Naming

Use the following format for assignment folders:

`task-[number]`

For this assignment:

`task-01`

Inside the folder, create:

`task-01/task-01.tex`

For future assignments, examples include:

- `task-02/task-02.tex`
- `task-03/task-03.tex`
- `task-04/task-04.tex`

You may create additional folders inside the assignment folder when necessary.
For example:

```
task-01/
|
+-- task-01.tex
|
+-- images/
|   +-- competition-aircraft.png
|
+-- references/
```

Keep all files related to an assignment inside its corresponding folder.

2.7 Branch Naming Convention

For each assignment, create a new branch.

Use the following format:

`task-[number]-[short-description]`

For this assignment:

`task-01-competition-research`

Examples for future assignments:

- `task-02-aircraft-analysis`
- `task-03-airfoil-research`
- `task-04-system-architecture`

The branch contains the changes for that assignment, while the assignment folder organizes the files within the repository.

2.8 Pull Request

After completing your assignment:

1. Commit your work to your assignment branch.
2. Push the branch to GitHub.
3. Open a Pull Request from your assignment branch into `main`.
4. Use a clear Pull Request title.
5. Wait for the assignment to be reviewed.

For Task 01, the Pull Request should look like:

```
task-01-competition-research
      ↓
    main
```

Do not merge your own Pull Request unless instructed to do so.
The training team will review your Pull Request and handle the merge.

3 Part 2 — Fixed-Wing Competition Research

Research one fixed-wing aircraft competition other than:

- Design/Build/Fly (DBF)
- ICMTC

Choose a competition that is relevant to aircraft design, aeromodelling, aerospace engineering, or autonomous flight.

3.1 Your Research Must Cover

1. Competition Overview

Include:

- Competition name
- Country
- Official website
- Short description

2. Deadlines

Include the important dates such as:

- Registration deadline
- Submission deadlines
- Competition dates
- Other important deadlines

3. Entry Fees

State the registration or entry fees and specify the currency.

4. Competition Format

Explain:

- Main mission or objective
- How the aircraft is operated
- How flights are conducted
- How teams are scored

5. Manual or Autonomous

Clearly state whether the competition uses:

- Manual control
- Autonomous operation
- Both

Explain how this affects the aircraft and its systems.

6. Aircraft Requirements

Identify the major requirements and restrictions affecting aircraft design.

7. Comparison With Lycans' Competitions

Compare the selected competition with DBF and ICMTTC.

Discuss differences in:

- Mission
- Aircraft requirements
- Payload
- Control format
- Scoring
- Major design constraints

4 Report Requirements

Your research must be submitted as a minimum **two-page LaTeX document**.

Use the provided Lycans LaTeX template.

Your report must contain:

- Clear sections and subsections
- At least one relevant figure
- At least one table
- References to your sources
- Properly formatted text
- Your own analysis and comparison

Do not submit screenshots of text, tables, or equations when they can be written directly in LaTeX.

5 Tools

For this task, use:

5.1 GitHub

GitHub will be used to store and submit your work.

<https://github.com/Lycans-Training-27>

5.2 Overleaf

Use Overleaf to write and compile your LaTeX document.

You do not need to install LaTeX locally for this task.

5.3 MiKTeX

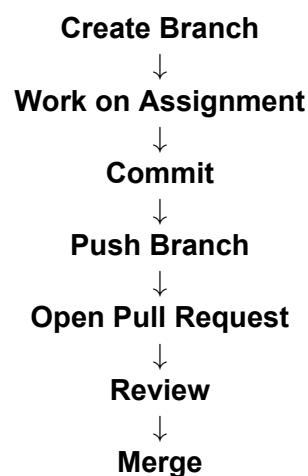
You do not need to install MiKTeX.

Research what MiKTeX is and how local LaTeX compilation differs from using Overleaf.

6 Submission

All assignments will follow the same GitHub workflow.

For every assignment:



6.1 Task 01 Workflow

For this task, follow these steps:

1. Create your personal training repository from the Lycans template.
2. Keep the original `assignment.tex` file unchanged.
3. Create a new branch: `task-01-competition-research`
4. Create a folder: `task-01/`
5. Create your assignment document: `task-01/task-01.tex`
6. Use the provided `assignment.tex` as a reference for formatting and LaTeX commands.

7. Research your selected competition.
8. Write your report in `task-01/task-01.tex`.
9. Compile and check your PDF using Overleaf.
10. Commit your changes.
11. Push your branch to GitHub.
12. Open a Pull Request from `task-01-competition-research` into `main`.
13. Wait for review.

6.2 Commit Message

Use a clear commit message describing your work.

Example:

```
Add competition research
```

Avoid unclear messages such as:

```
update      final      changes
```

6.3 Pull Request Title

Use the same naming style for your Pull Request.

Example:

```
Task 01 -- Fixed-Wing Competition Research
```

Your Pull Request should clearly communicate what assignment you are submitting.

6.4 Important

Do not create a new repository for every assignment.

You will keep using your personal training repository.

For each new assignment:

- Create a new branch.
- Complete the assignment on that branch.
- Commit your work.
- Push the branch.
- Open a Pull Request into `main`.

This workflow will be used for all future training assignments.

Research well. Document clearly. Submit professionally.